Cell Thawing

C03 | ver 1.1 | Revised 2026-10-03

SUSTech-BIO 2026 iGEM Team

Purpose

Recover cryopreserved adherent mammalian cells by rapid thawing, dilution with warm complete medium, centrifugation and reseeding. Use this general workflow for routine cell lines that tolerate post-thaw centrifugation; retain any cell-specific recovery requirements.

Materials & Equipment

Item	Supplier / identifier	Use / specification
Cryopreserved cells	Identified cell stock	Cell line, passage and frozen suspension volume known
Complete growth medium	Cell-line-specific	Correct basal medium and supplements; prewarmed to 37°C
Sterile centrifuge tubes	Sterile laboratory consumables	15 mL tubes for the 1 mL vial example
Tissue-culture-treated vessel	Laboratory stock	Suitable plate, dish or flask; coating if required
Sterile pipettes and tips	Axygen tips; laboratory supplies	Gentle transfer, dilution and resuspension
70% ethanol	Sangon Biotech ethanol; 70% working solution	External vial decontamination
Water bath; cell culture cabinet	Laboratory equipment	37°C bath; use a Class II biosafety cabinet where required for the cell line
Centrifuge; inverted microscope	Eppendorf centrifuge; laboratory equipment	Low-speed cell pelleting and culture inspection
Humidified incubator	Laboratory equipment	Temperature and CO2 matched to cell line and medium

Before You Start

Prewarm sufficient complete medium to 37°C before retrieving the frozen vial. Prepare both the dilution volume (five times the frozen suspension volume) and fresh medium for resuspension and culture. Do not start thawing while the medium is still warming.

Set the water bath to 37°C. Label the receiving tube and culture vessel, prepare any required coating, and have the centrifuge and culture cabinet ready. Keep the cell stock frozen until these

preparations are complete.

Select the vessel area from the expected viable cell yield and the cell line's recommended recovery density; avoid spreading a low-yield vial over too large an area. Match the medium and incubator atmosphere to the cell line. [1, 2]

Procedure

A. Rapid thawing

1. Retrieve one cryovial and promptly place its lower portion in the 37°C water bath. Keep the cap and seal above the water. Follow the laboratory's cryogenic-handling procedure when removing vials from liquid nitrogen.

2. Keep the vial moving with continuous, gentle rocking or swirling throughout thawing. Watch the contents: as soon as only a small piece of ice remains, remove the vial and begin transferring it to the prepared culture cabinet. Do not wait in the bath for the last ice crystal to disappear.

3. Wipe the vial exterior with 70% ethanol before opening it in the cabinet. Work promptly and aseptically; allow the small remaining ice fragment to finish melting during handling rather than prolonging water-bath exposure. [1]

B. Dilute and remove freezing medium

4. Transfer the thawed suspension into a sterile centrifuge tube. Gradually add prewarmed complete medium equal to five times the original frozen suspension volume, mixing gently as it is added. Avoid a single forceful addition or vigorous pipetting.

Quantity	General rule	Example
Original frozen suspension	V	1 mL
Warm medium to add	5 × V	5 mL
Volume before centrifugation	Approximately 6 × V	Approximately 6 mL

5. Balance the tubes and centrifuge at 200 × g for 5 min at room temperature as a general starting condition. Use the cell supplier's specified condition if it differs. This setting is selected from the general 200 × g, 5-10 min guidance. [1]

6. Locate the cell pellet and carefully aspirate the supernatant in the cabinet. Keep the tip away from the pellet; remove the diluted freezing medium without dislodging the cells.

C. Resuspend, plate and recover

7. Gently loosen and resuspend the pellet in fresh, prewarmed complete medium. Use enough medium to disperse the cells and then adjust to the working volume of the selected vessel. Avoid frothing or repeated forceful pipetting.

8. Transfer the suspension to the prepared culture vessel. Distribute cells evenly by gentle back-and-forth and side-to-side movement, then place the vessel flat in the incubator.

9. Incubate under the cell line's established conditions; 37°C with 5% CO2 is common for mammalian cells grown in compatible bicarbonate-buffered media. Leave the vessel undisturbed during initial attachment. [2]

10. Inspect the culture under an inverted microscope the following day, approximately 24 h after thawing. Record attachment, morphology, debris and medium appearance. Once the cells have attached, gently replace the medium if needed to remove floating debris, following the cell line's recovery instructions. [2]

p.s.

Finish warming the medium and preparing the receiving tube before taking cells out of storage. Thaw one vial at a time when working alone so dilution and centrifugation can follow promptly.

Continuous movement in the bath means gentle rocking or swirling, not vigorous shaking. The small remaining ice fragment is the cue to leave the bath; a fixed thawing time should not override that visible endpoint.

Add five volumes of warm medium to one volume of cell suspension. For a 1 mL vial, add 5 mL: do not merely top up to a final volume of 5 mL.

A small pellet may be translucent and loosely attached. Watch its position while aspirating and resuspend gently; avoid disturbing newly attached cells with repeated early handling.

Expected Outcome

Recovering adherent cells should progressively attach, regain their characteristic morphology and resume growth. Some floating cells and debris can occur after thawing; judge recovery from subsequent attachment and growth rather than demanding a completely debris-free culture immediately. [3] Record the stock identity, vessel, medium, centrifugation setting and next-day observations.

Troubleshooting

Observed problem	Check first	Suggested action
Thawing is unexpectedly slow	Bath temperature, vial contact with water and continuous movement	Confirm 37°C and maintain gentle movement. Use the small-ice endpoint; do not compensate by overheating the bath.
Pellet is difficult to see	Starting cell quantity, tube orientation and centrifuge setting	Inspect before aspiration and verify × g rather than assuming an rpm equivalent. Do not discard the suspension or automatically increase force.
Poor attachment or slow recovery	Correct medium, vessel/coating, density and thaw-to-plating delay	Correct culture conditions; use the recommended recovery density and minimize handling. Review freezing and storage history if recovery remains poor. [1, 3]
Many floating cells or debris	Whether attached cells are recovering; signs of progressive deterioration	Monitor the attached population. After attachment, change medium gently when needed; persistent deterioration warrants checking stock quality and contamination. [3]
Cloudy medium or unusual particles	Microscopy, aseptic handling and shared reagents	Separate the suspect culture and investigate contamination under the laboratory procedure. Do not rely on a medium change to resolve contamination. [2]

Reference

[1] Thermo Fisher Scientific / Gibco. Thawing Cells: general protocol and troubleshooting. Centrifugation guidance: approximately 200 × g for 5-10 min, adjusted to cell type. Accessed 2026-10-03. Open source

[2] ATCC. Animal Cell Culture Guide: culture conditions, recovery of cryopreserved cells, and contamination. Accessed 2026-10-03. Open source

[3] ATCC. Reviving Cells: post-thaw recovery and delayed recovery of some cell lines. Accessed 2026-10-03. Open source